

Autonomous Supply Chain Intelligence Pipeline

Project Blueprint, Technical Architecture & Monetization Strategy

1. Executive Summary & Value Proposition

This project bridges Supply Chain Management (SCM) domain expertise with Agentic AI orchestration, short-form video generation, and cloud automation. The system monitors real-time ocean freight trends, parses operational news via CrewAI agents, generates executive summaries using the Gemini API, renders video assets via ZSky/Veo, and publishes live feeds to social media and a custom web dashboard hosted on Vercel at \$0 server cost.

2. Target SCM Niches

- * Global Shipping & Port Delays: Real-time vessel backlogs and ocean freight rate changes.
- * Logistics & Warehouse AI: Daily briefs on warehouse robotics, WMS software, and route planning.
- * Procurement & Safety Stock Strategy: Concise tips on EOQ, safety stock formulas, and risk.
- * Sustainability & Circular Logistics: Scope 3 carbon reduction and green fleet updates.

3. End-to-End Technology Stack (\$0 Operating Overhead)

Layer	Selected Tool	Role in Pipeline
Design & UI	Google Stitch & Websiteprompts	Generates web UI code & dashboard layout
Brand DNA	Google Pomelli	Establishes visual brand identity and styles
AI Code Assistant	Google Antigravity / Gemini	AI-guided code generation inside VS Code
AI Agent Engine	CrewAI + Gemini 2.5 API	Multi-agent research & text summarization
Video Generation	ZSky / Google Veo API	Programmatic vertical AI video rendering
Browser Scraping	Bardeen AI / Feedparser	Data extraction for non-API web sources
Web Dashboard	Next.js / Streamlit (Vercel)	Live web dashboard displaying logs/charts
Cloud Automation	GitHub Actions	Runs daily scheduled pipeline at \$0 cost

4. Monetization Strategy

1. B2B Sponsored Newsletter: Automate a weekly digest via Beehiiv. Sell banner slots once active.
2. Affiliate Lead Generation: Integrate referral links for SCM software, courses, and tools.
3. High-Ticket B2B Consulting: Use live website logs and GitHub code as proof-of-work.

5. 4-Week Implementation Roadmap

- Week 1: Setup VS Code, Gemini API Key, and write local CrewAI agent scripts.
- Week 2: Generate UI layouts via Google Stitch and deploy Next.js/Streamlit to Vercel.
- Week 3: Connect video APIs (ZSky/Veo) and schedule daily runs on GitHub Actions.
- Week 4: Embed newsletter signups, publish GitHub repository, and update resume.